PIPES & CISTERNS

Placement Aptitude Test

Total Questions: 50

Time Allowed: 50 Minutes

Marking Scheme: +1 for each correct answer, -0.25 for each incorrect answer

Section-wise Break-up

Section	Topic	Questions
A	Two-Pipe Basics	8
B	Three-or-More Pipes Together	7
C	Inlet & Outlet Combined	10
D	Leaks in the Cistern	8
E	Alternate Opening / Sequenced Pipes	8
F	Advanced & Mixed Applications	9
	Total	**50**

Instructions:

1. This test contains 50 multiple-choice questions across 6 sections.
2. Each question carries 1 mark. There is a negative marking of 0.25 for each wrong answer.
3. There is no negative marking for unattempted questions.
4. Use of calculators is not permitted unless specified.
5. The answer key is provided at the end of this document.

Two-Pipe Basics

1. Pipe A can fill a cistern in 12 hours and pipe B can fill it in 18 hours. If both pipes are opened together, in how much time will the cistern be full?

(A) 7 h 12 min (B) 7 h 30 min

(C) 8 h (D) 6 h 40 min

2. Two pipes can fill a tank in 20 hours and 30 hours respectively. If both are opened together, find the time taken to fill the tank.

(A) 10 h (B) 12 h

(C) 15 h (D) 25 h

3. Pipe A can fill a tank in 6 hours while pipe B can empty the same full tank in 8 hours. If both are opened together on an empty tank, in how many hours will it be filled?

(A) 18 h (B) 20 h

(C) 24 h (D) 28 h

4. Two pipes can fill a cistern in 10 minutes and 15 minutes respectively. Find the time taken to fill the cistern if both pipes are opened together.

(A) 5 min (B) 7.5 min

(C) 12 min (D) 6 min

5. Pipe A alone fills a tank in 24 hours and pipe B alone fills it in 32 hours. Working together, how long will they take to fill the tank?

(A) 13 5/7 h (B) 13 3/7 h

(C) 14 h (D) 12 6/7 h

6. A pipe fills 3/4 of a tank in 15 minutes. Another pipe alone can fill the full tank in 30 minutes. If both are opened together, how long will they take to fill the empty tank?

(A) 10 min (B) 12 min

(C) 15 min (D) 18 min

7. Two pipes can fill a tank in 12 minutes and 16 minutes respectively. If both pipes are opened together, in what time will the tank be full?

(A) 6 3/7 min (B) 7 1/7 min

(C) 6 6/7 min (D) 7 3/7 min

8. Pipe A is thrice as fast as pipe B. If both together can fill a cistern in 36 minutes, find the time in which pipe A alone can fill the cistern.

(A) 36 min (B) 42 min

(C) 54 min (D) 48 min

Three-or-More Pipes Together

9. Three pipes A, B and C can fill a cistern in 6 hours, 10 hours and 15 hours respectively. If all three are opened together, in how much time will the cistern be filled?

(A) 3 h (B) 4 h

(C) 5 h (D) 6 h

10. Four identical pipes can each fill a tank in 20 hours. If all four are opened together, how long will they take to fill the tank?

(A) 4 h (B) 5 h

(C) 6 h (D) 8 h

11. Pipes A, B and C can fill a tank in 10 hours, 15 hours and 30 hours respectively. If all three pipes are opened simultaneously, find the time to fill the tank.

(A) 4 h (B) 6 h

(C) 5 h (D) 7.5 h

12. Pipe C alone fills a cistern in 84 minutes. Pipe B is twice as fast as C, and pipe A is twice as fast as B. If all three pipes are opened together, find the time to fill the cistern.

(A) 10 min (B) 14 min

(C) 16 min (D) 12 min

13. Pipes A, B and C can fill a cistern in 24 minutes, 32 minutes and 48 minutes respectively. All three are opened together, but pipe A is closed after 8 minutes. Find the total time taken to fill the cistern.

(A) 12.8 min (B) 11.6 min

(C) 12.4 min (D) 13.2 min

14. Pipes A, B and C can fill a cistern in 6 hours, 8 hours and 12 hours respectively. Pipe A is opened first; pipe B is opened 1 hour later; pipe C is opened 1 hour after that (i.e., 2 hours after A). Find the total time taken to fill the cistern from the moment A was opened.

(A) 3 h 20 min (B) 3 h 26 2/3 min

(C) 3 h 30 min (D) 3 h 40 min

15. Pipes A and B together can fill a tank in 12 hours; B and C together in 15 hours; A and C together in 20 hours. Find the time taken by all three pipes together to fill the tank.

(A) 8 h (B) 9 h

(C) 10 h (D) 12 h

Inlet & Outlet Combined

16. An inlet pipe can fill a tank in 12 hours; an outlet pipe can empty the full tank in 20 hours. If both are opened together on an empty tank, in how long will the tank be filled?

(A) 24 h (B) 28 h

(C) 32 h (D) 30 h

17. An inlet pipe fills a tank in 8 hours; an outlet pipe empties the full tank in 12 hours. Both pipes are opened together on an empty tank. Find the time taken to fill it.

(A) 24 h (B) 20 h

(C) 22 h (D) 26 h

18. A cistern has two inlet pipes A and B, and one outlet pipe C. A alone fills it in 10 hours, B alone in 15 hours, and C alone empties it in 6 hours. If all three pipes are opened together on an empty cistern, what happens?

(A) Fills in 30 h (B) Never fills — level stays unchanged

(C) Fills in 60 h (D) Empties in 30 h

19. A tank has an inlet that fills it in 6 hours and an outlet that empties it in 9 hours. If both pipes are opened together on an empty tank, how long will it take to fill?

(A) 15 h (B) 16 h

(C) 18 h (D) 20 h

20. Two inlet pipes A and B can fill a tank in 12 hours and 15 hours respectively; an outlet pipe C can empty the full tank in 10 hours. If all three are opened together on an empty tank, find the time taken to fill it.

(A) 18 h (B) 24 h

(C) 25 h (D) 20 h

21. Pipe A fills a tank in 5 hours; pipe B empties it in 9 hours. The tank is already 2/5 full. If both pipes are now opened together, in how long will the tank become completely full?

(A) 6 h 45 min (B) 5 h 15 min

(C) 6 h (D) 7 h 30 min

22. Pipe A can fill a cistern in 10 hours; pipe B can empty it in 15 hours. Both are opened together on an empty cistern. After 6 hours, pipe B is closed. Find how much longer pipe A alone takes to fill the remaining part.

(A) 6 h (B) 8 h

(C) 7 h (D) 9 h

23. A tank can be filled by pipe A in 12 hours and emptied by pipe B in 20 hours. The tank is already 1/4 full. If both pipes are opened together, after how long will the tank be completely full?

(A) 20 h (B) 21 h

(C) 22 h 30 min (D) 24 h

24. Two inlet pipes A and B can fill a tank in 20 hours and 25 hours respectively; an outlet pipe C can empty it in 15 hours. If all three pipes are opened together on an empty tank, find the time to fill it.

(A) 40 h (B) 45 h

(C) 48 h (D) 42 6/7 h

25. Taps A and B can fill a tank in 15 hours and 20 hours respectively. A and B are opened together, and after 4 hours, outlet tap C (which can empty the full tank in 12 hours) is also opened. Find the total time taken to fill the tank.

(A) 20 h (B) 16 h

(C) 18 h (D) 22 h

Leaks in the Cistern

26. A pipe can fill a tank in 6 hours. Because of a leak in the bottom, it takes 8 hours to fill the tank. If the tank is full, how long will the leak alone take to empty it?

(A) 20 h (B) 24 h

(C) 28 h (D) 32 h

27. A pipe can fill a cistern in 10 hours in the absence of a leak. Because of a leak, it now takes 2 hours more. Find the time the leak alone would take to empty the full cistern.

(A) 50 h (B) 55 h

(C) 60 h (D) 65 h

28. A leak in the bottom of a tank can empty a full tank in 12 hours. An inlet pipe fills water at 6 litres/minute. When the tank is full, the inlet is opened, and despite this, the leak empties the tank in 20 hours. Find the capacity of the tank.

(A) 9600 L (B) 10200 L

(C) 11400 L (D) 10800 L

29. Two pipes A and B can fill a tank in 12 hours and 15 hours respectively. There is also a leak which can empty the full tank in 40 hours. If A, B and the leak all act together on an empty tank, find the time to fill it.

(A) 8 h (B) 6 h

(C) 7 h (D) 9 h

30. A pump can fill a tank in 3 hours. Because of a leak, it takes 3 hours 30 minutes to fill the tank. If the tank is full, find the time the leak alone would take to empty it.

(A) 18 h (B) 21 h

(C) 24 h (D) 27 h

31. Two leaks L1 and L2 can empty a full tank in 20 hours and 30 hours respectively. An inlet pipe alone can fill the tank in 10 hours. If the inlet and both leaks act together on an empty tank, find the time to fill it.

(A) 50 h (B) 55 h

(C) 60 h (D) 65 h

32. A tank is filled by pipe A alone in 8 hours. Pipe A together with a leak fills the tank in 10 hours. If the tank is full and only the leak is opened, how long will it take to empty half the tank?

(A) 15 h (B) 18 h

(C) 24 h (D) 20 h

33. A cistern is filled by an inlet pipe in 4 hours. Because of two leaks — one at the bottom and one at the side — it actually takes 6 hours to fill. If the bottom leak alone would empty the full tank in 20 hours, find the time the side leak alone would take to empty it.

(A) 30 h (B) 24 h

(C) 28 h (D) 32 h

Alternate Opening / Sequenced Pipes

34. Pipe A can fill a tank in 12 hours and pipe B in 24 hours. They are opened alternately for 1 hour each, starting with A. Find the total time taken to fill the tank.

(A) 14 h (B) 16 h

(C) 15 h (D) 18 h

35. Pipe A can fill a tank in 6 hours and pipe B in 8 hours. They are opened alternately for 1 hour each, starting with A. Find the total time taken to fill the tank.

(A) 6 h 30 min (B) 7 h

(C) 6 h 45 min (D) 7 h 15 min

36. Pipes A and B can fill a cistern in 10 hours and 15 hours respectively. Both are opened together, but after 3 hours, pipe A is turned off. Find the total time taken to fill the cistern.

(A) 9 h (B) 9 h 30 min

(C) 10 h (D) 10 h 30 min

37. Pipe A can fill a cistern in 12 hours. It is opened alone for 4 hours; then pipe B (which alone fills the cistern in 8 hours) is also opened. Find the total time taken to fill the cistern from the start.

(A) 7 h 12 min (B) 6 h 40 min

(C) 7 h (D) 7 h 30 min

38. Pipe A can fill a cistern in 20 hours. An outlet pipe B can empty the full cistern in 30 hours. Pipe B is opened 5 hours after pipe A. Find the total time taken to fill the cistern.

(A) 45 h (B) 50 h

(C) 48 h (D) 52 h

39. Pipe A can fill a tank in 6 hours and pipe B in 10 hours. Both are opened together, but pipe B is closed 2 hours before the tank becomes full. Find the total time taken to fill the tank.

(A) 4 h (B) 4 h 15 min

(C) 4 h 30 min (D) 4 h 45 min

40. A tank is already half full. Pipe A (which fills the full tank in 12 hours) and pipe B (which empties the full tank in 18 hours) are both opened. In how much time will the tank become completely full?

(A) 16 h (B) 20 h

(C) 24 h (D) 18 h

41. Pipe A alone can fill a cistern in 5 hours. Pipes A and B together can fill it in 3 hours. If pipe B alone is used to fill only 2/5 of the cistern, how long will it take?

(A) 3 h (B) 2 h

(C) 2.5 h (D) 3.5 h

Advanced & Mixed Applications

42. A cistern has two inlet pipes that can fill it in 12 hours and 20 hours, and one outlet pipe that can empty the full cistern in 15 hours. If all three pipes are opened together on an empty cistern, find the time taken to fill it.

(A) 12 h (B) 15 h

(C) 18 h (D) 20 h

43. A pipe can empty a cistern in 40 minutes. A second pipe with twice the diameter of the first (and hence four times the flow rate, since flow rate is proportional to the square of the diameter) is also opened. Find the time taken by both pipes together to empty the cistern.

(A) 6 min (B) 7 min

(C) 8 min (D) 10 min

44. Two pipes can fill a tank in 15 hours and 20 hours respectively. Both are opened together, but due to a crack, only 5/6 of the water actually supplied stays in the tank (1/6 leaks out continuously). Find the effective time taken to fill the tank.

(A) 10 h (B) 10 4/7 h

(C) 11 h (D) 10 2/7 h

45. Pipe A can fill a tank in x hours and pipe B can fill it in (x+6) hours. If both together fill the tank in 4 hours, find the time pipe A alone takes to fill the tank.

(A) 6 h (B) 4 h

(C) 8 h (D) 10 h

46. Pipe A is 60% as efficient as pipe B. If pipe B alone can fill a tank in 25 hours, find the time taken by A and B together to fill the tank.

(A) 14 h (B) 15 5/8 h

(C) 15 h (D) 16 h

47. Pipe A can fill a tank 4 times as fast as pipe B, and A takes 30 minutes less than B to fill the tank. Find the time in which A and B together can fill the tank.

(A) 6 min (B) 7 min

(C) 8 min (D) 9 min

48. Pipes A and B can fill a cistern in 8 hours and 12 hours respectively. An outlet pipe C is also connected. When all three pipes are opened together, the cistern fills in 24 hours. Find the time in which pipe C alone can empty the full cistern.

(A) 4 h (B) 5 h

(C) 7 h (D) 6 h

49. Pipe A alone can fill a tank in 6 hours and pipe B alone in 4 hours. Pipe A is opened at 2:00 PM and pipe B is opened at 3:00 PM. At what time will the tank be completely full?

(A) 5:00 PM (B) 4:30 PM

(C) 5:30 PM (D) 6:00 PM

50. A swimming pool has 3 identical inlet pipes and 2 identical outlet pipes. Each inlet alone fills the pool in 24 hours; each outlet alone empties it in 36 hours. If all 5 pipes are opened together on an empty pool, find the time taken to fill it.

(A) 12 h (B) 14 h 24 min

(C) 13 h 12 min (D) 15 h 36 min

ANSWER KEY

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	A	11	C	21	A	31	C	41	A
2	B	12	D	22	B	32	D	42	B
3	C	13	A	23	C	33	A	43	C
4	D	14	B	24	D	34	B	44	D
5	A	15	C	25	A	35	C	45	A
6	B	16	D	26	B	36	D	46	B
7	C	17	A	27	C	37	A	47	C
8	D	18	B	28	D	38	B	48	D
9	A	19	C	29	A	39	C	49	A
10	B	20	D	30	B	40	D	50	B